

Samson Vision

Visual-to-text bridge for AI agents

Structured **SVP** (SAMSON_VISION_PACK) for text-only LLMs — screenshots, documents, and UI without native vision APIs.

Early release — honest scope. Samson Vision is **not** a replacement for native multimodal vision. It is a cheap, auditable bridge: structured text so text-only LLMs can reason over screenshots and UI. Benchmarks use **6/6 binary signals**, not 100% visual quality claims.

Use Cases

- **Web screenshots** — Capture page structure, headlines, and navigation for agent reasoning.
- **Dashboards & analytics** — Extract KPIs, chart labels, and layout from BI screens.
- **Documents & OCR** — Scanned PDFs and forms — readable text plus spatial layout.
- **Form & UI review** — Fields, buttons, states — coordinate-aware for automation agents.
- **Error screens & logs** — Stack traces and alert dialogs as structured, searchable text.
- **Visual QA / CI** — Versionable SVP diffs for regression checks without vision APIs.

What SVP Delivers

- OCR text with position & confidence (Tesseract)
- Layout map, visual hierarchy & object regions (OpenCV)
- ASCII representation (8 styles) + color map + density bands
- Pixel coordinates (0–100 scale) for click/automation hints
- Anti-hallucination rules — auditable, deterministic pipeline (0% AI at generation)

Integration Modes

Mode	Stack	Best for
A — Text only	SVP + text LLM	Cheap agents, OCR/layout, CI diffs
B — SVP + subagent	Orchestrator reads SVP → vision subagent validates	UI review, accessibility, regressions
C — Native vision	Multimodal model on raw image	Photos, logos, fine detail (fidelity > cost)

Costs (per query, not monthly)

SVP generation is **local & free** (compute only) — no per-image API. Interpretation uses standard text tokens (~1,700 in + 400 out per query).

Approach	Example	Cost / query	Notes
SVP gen only	samson-vision CLI	\$0	Local: numpy + OpenCV + Tesseract
SVP + text LLM	MiniMax-M2.1	\$0.0008	6/6 signals on benchmark
SVP + text LLM	GPT-4o-mini	~\$0.0005	Solid text interpreter

Approach	Example	Cost / query	Notes
Direct vision	GPT-4o	~\$0.008	Native multimodal — 4–10× costlier
Direct vision	MiniMax-M3	~\$0.003	Better for photos; higher per-call cost

Text agent + cached SVP vs. sending every image to a vision model: fewer multimodal calls, predictable token billing.

Install

```
pip install "samson-vision[dev] @ git+https://github.com/TonyRiera/samson-vision.git@v0.3.3"
```

github.com/TonyRiera/samson-vision